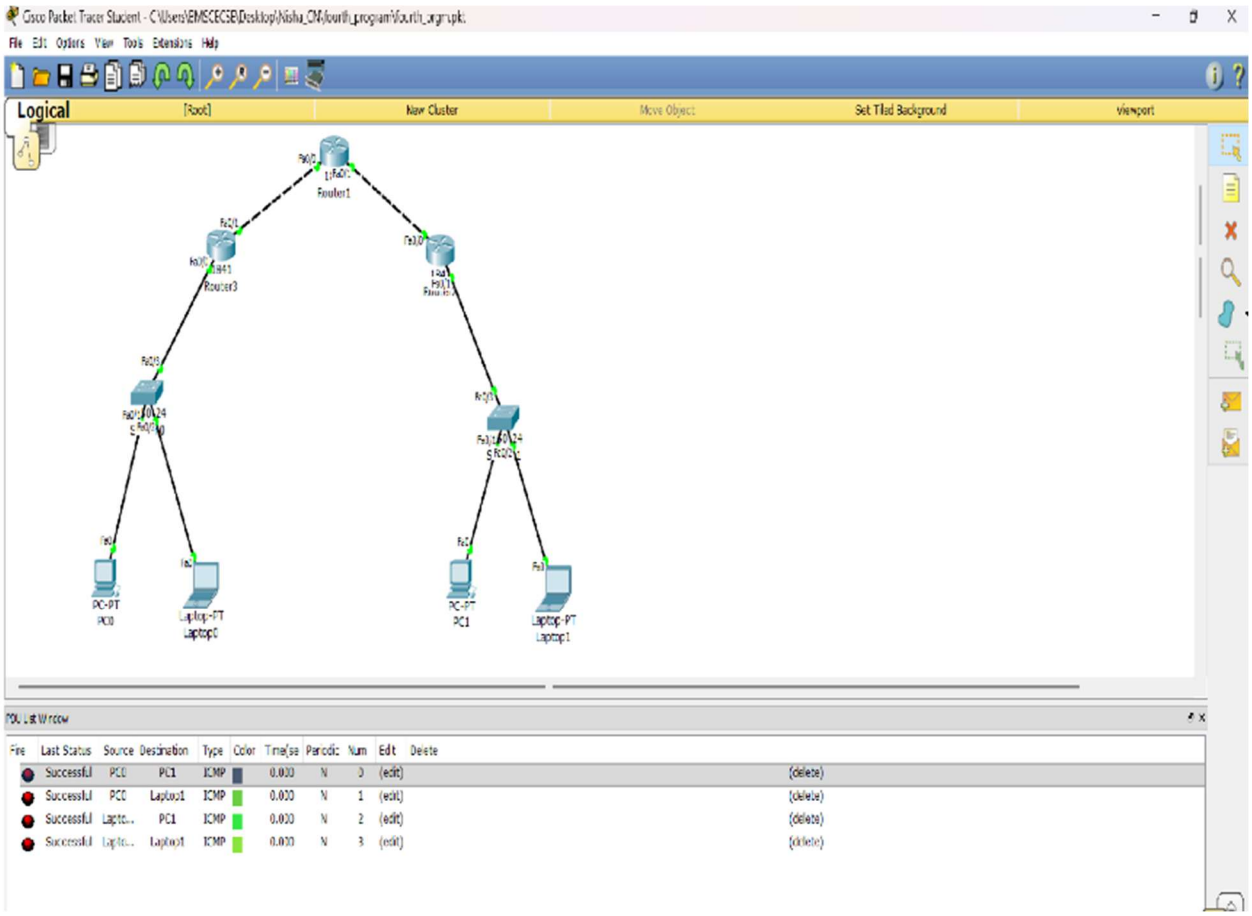
Lab program 04

4. How to configure Static Route and Destination Route to the Routers




```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
Router(config)#ip route 40.0.0.0 255.0.0.0 30.0.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
```

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 20.0.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
```

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 30.0.0.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
```

```
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 20.0.0.2 to network 0.0.0.0

C    10.0.0.0/8 is directly connected, FastEthernet0/0
C    20.0.0.0/8 is directly connected, FastEthernet0/1
S*   0.0.0.0/0 [1/0] via 20.0.0.2
Router>
```

```
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
C    20.0.0.0/8 is directly connected, FastEthernet0/0
C    30.0.0.0/8 is directly connected, FastEthernet0/1
S    40.0.0.0/8 [1/0] via 30.0.0.2
Router>
```

```
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 30.0.0.1 to network 0.0.0.0

C    30.0.0.0/8 is directly connected, FastEthernet0/0
C    40.0.0.0/8 is directly connected, FastEthernet0/1
S*   0.0.0.0/0 [1/0] via 30.0.0.1
Router>
```

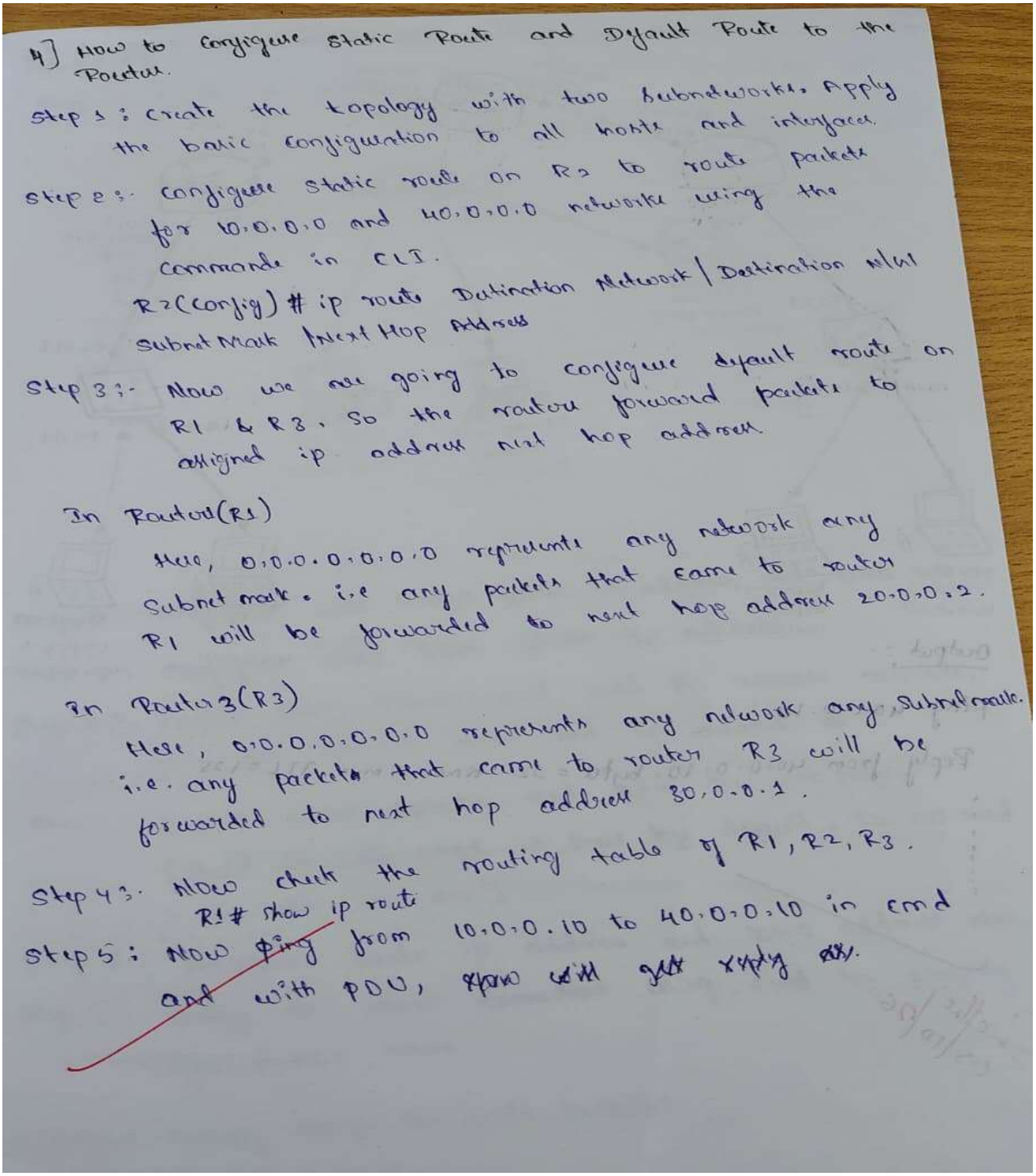
```
PC>
PC>ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:

Reply from 40.0.0.10: bytes=32 time=191ms TTL=125
Reply from 40.0.0.10: bytes=32 time=111ms TTL=125
Reply from 40.0.0.10: bytes=32 time=156ms TTL=125
Reply from 40.0.0.10: bytes=32 time=127ms TTL=125

Ping statistics for 40.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 111ms, Maximum = 191ms, Average = 146ms
```

Observation :



4] How to configure Static Route and Default Route to the Router.

Step 1 : Create the topology with two subnetworks, Apply the basic configuration to all hosts and interfaces.

Step 2 :- Configure static route on R2 to route packets for 10.0.0.0 and 40.0.0.0 networks using the commands in CLI.

R2(config) # ip route Destination Network / Destination N/W Subnet Mask Next Hop Address

Step 3 :- Now we are going to configure default route on R1 & R3. So the router forward packets to assigned ip address next hop address.

In Router(R1)

Here, 0.0.0.0 0.0.0.0 represents any network any subnet mask. i.e any packets that came to router R1 will be forwarded to next hop address 20.0.0.2.

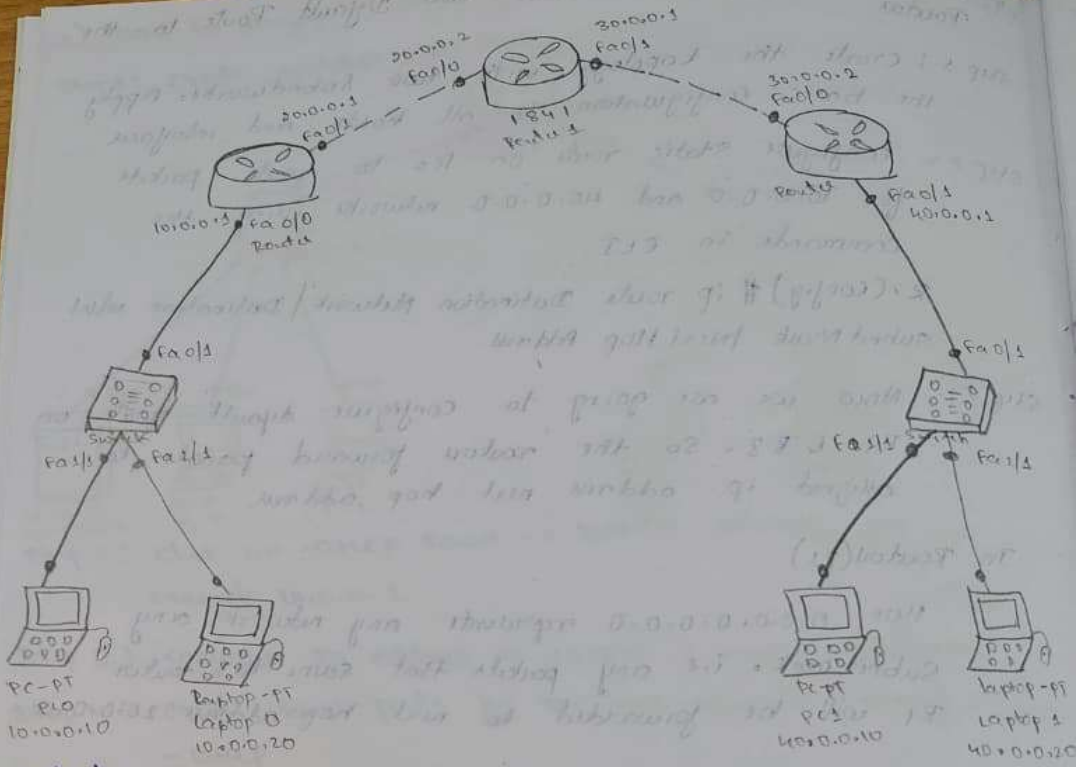
In Router3(R3)

Here, 0.0.0.0 0.0.0.0 represents any network any subnet mask. i.e. any packets that came to router R3 will be forwarded to next hop address 30.0.0.1.

Step 4 :- Now check the routing table of R1, R2, R3.

R1 # show ip route

Step 5 : Now ping from 10.0.0.10 to 40.0.0.10 in cmd and with PDU, ping will get replay.



Output :-

ping 40.0.0.10

Reply from 40.0.0.10 byte = 32 time = 15 min TTL = 125

.....

Scanned
05/10/26